

LeetCode Notes: Find Smallest Set of Vertices

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Key Observations & Explanation

- basic si observation he bas
- jis node ki indegree ≥ 1 he wo kahi na kahi se reach ho jaayega
- jis node ki indegree $= 0$ he usko include krna majburi he kyunki wo unreachable he

C++ Solution

```
class Solution {
public:
    vector<int> findSmallestSetOfVertices(int n, vector<vector<int>>& edges) {

        //basic si observation he bas
        //jis node ki indegree $\geq 1$  he wo kahi na kahi se reach ho jaayega
        //jis node ki indegree $= 0$  he usko include krna majburi he kyunki
        wo unreachable he

        vector<vector<int>> adj(n);
        vector<int> indeg(n, 0);

        for(auto &it : edges) {
            int u = it[0];
            int v = it[1];
```

```
        adj[u].push_back(v);
        indeg[v]++;
    }

    vector<int> ans;
    for(int i = 0; i < n; i++) {
        if(indeg[i] == 0) {
            ans.push_back(i);
        }
    }

    return ans;
}

};
```